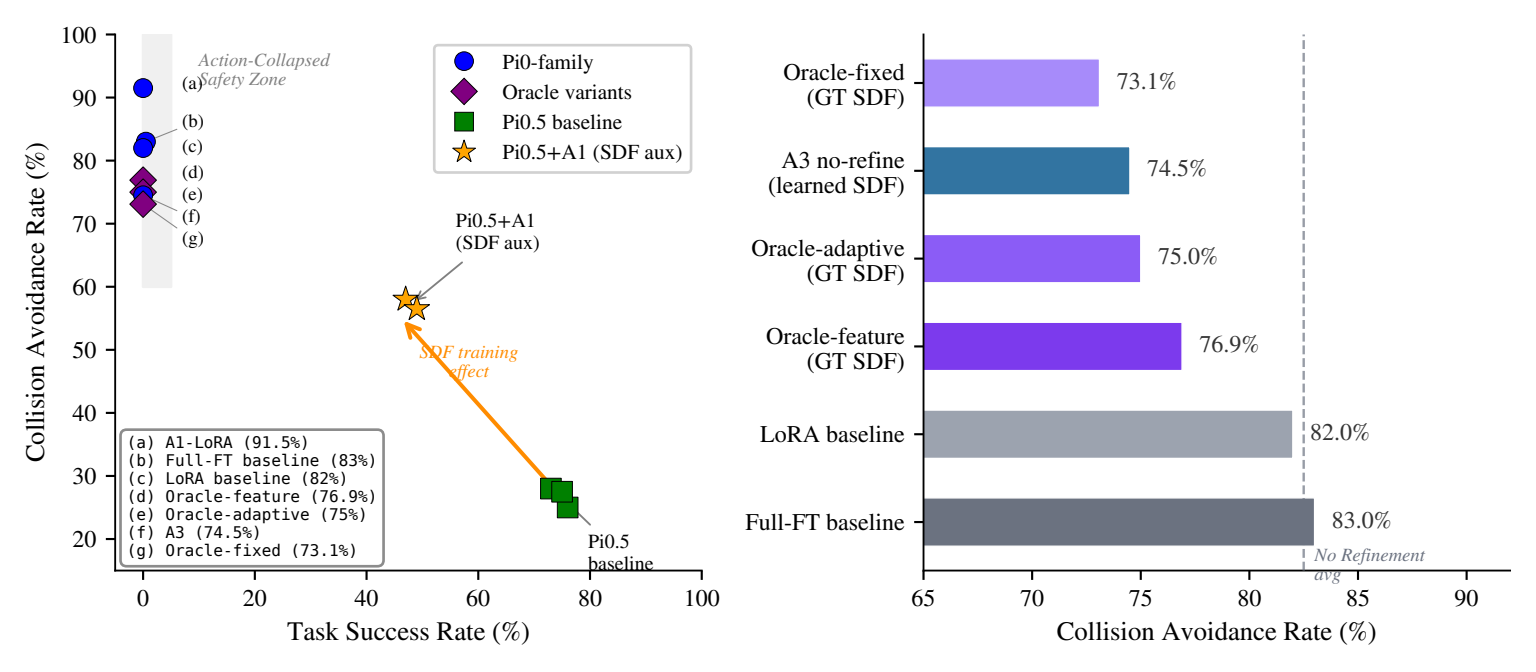
Collision Avoidance Rate (%)
Task Success Rate (%)
Action-Collapsed Safety Zone
Pi0-family
Oracle variants
Pi0.5 baseline
Pi0.5+A1 (SDF aux)
(a)
(b)
(c)
(d)
(e)
(f)
(g)
Pi0.5+A1 (SDF aux)
SDF training effect
Pi0.5 baseline
(a) A1-LoRA (91.5%)
(b) Full-FT baseline (83%)
(c) LoRA baseline (82%)
(d) Oracle-feature (76.9%)
(e) Oracle-adaptive (75%)
(f) A3 (74.5%)
(g) Oracle-fixed (73.1%)
Oracle-fixed (GT SDF)
73.1%
A3 no-refine (learned SDF)
74.5%
Oracle-adaptive (GT SDF)
75.0%
Oracle-feature (GT SDF)
76.9%
LoRA baseline
82.0%
Full-FT baseline
83.0%
No Refinement avg
Collision Avoidance Rate (%)